System Dynamics and Control Objectives

Understanding the drone's physical characteristics and dynamic response is essential for setting effective control objectives.

Alt text: *A human operating drone*

1. Analysing Physical Characteristics

This involves understanding the drone's mass, inertia, thrust capabilities, and aerodynamic properties.

Example: A lightweight racing drone will have different control requirements compared to a heavy-lift drone used for cargo delivery. The racing drone requires quick responses and minimal lag, while the cargo drone needs stable and precise control to handle heavy loads.

2. Dynamic Response

This refers to how the drone reacts to control inputs and external disturbances. Analysing dynamic responses helps in designing control systems that can handle real-world conditions.

Example: Test how the drone responds to wind gusts or sudden changes in direction. A well-tuned PID controller will adjust the motor speeds to counteract these disturbances and maintain stable flight.

3. Setting Control Objective

Control objectives define the desired performance criteria for the drone, such as altitude stabilisation and rotational motion control (pitch, roll, yaw).

Example: For altitude stabilisation, the objective might be to maintain a set altitude within a tolerance of ±1 meter, even in varying weather conditions. For rotational motion control, the objective could be to achieve smooth and precise turns without overshooting.